

Nama Puput Novitasari

Nim 2511102441336

Kelas A

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== -- Judul dan pemisah bagian
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-- DATABASE KASIR TOKO
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-- FITUR: Manajemen produk, transaksi, user, supplier, dan laporan
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-- BAGIAN 1: MANAJEMEN DATABASE
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DROP DATABASE IF EXISTS toko_kasir; -- Menghapus database jika sudah ada
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CREATE DATABASE toko_kasir; -- Membuat database baru
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USE toko_kasir; -- Menggunakan database
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-- BAGIAN 2: MEMBUAT TABEL ROLES
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CREATE TABLE Roles (
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    role_id INT PRIMARY KEY AUTO_INCREMENT, -- ID role
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    role_name VARCHAR(50) UNIQUE NOT NULL, -- Nama role
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        level INT DEFAULT 1 -- Level akses
    );

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-- BAGIAN 3: MEMBUAT TABEL USERS
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CREATE TABLE Users ( -- Tabel data pengguna
    user_id INT PRIMARY KEY AUTO_INCREMENT, -- ID user

    username VARCHAR(50) UNIQUE NOT NULL, -- Username

    password VARCHAR(255) NOT NULL, -- Password

    full_name VARCHAR(100) NOT NULL, -- Nama lengkap

    role_id INT NOT NULL, -- Relasi ke Roles

    is_active BOOLEAN DEFAULT 1, -- Status aktif

    last_login DATETIME, -- Login terakhir

    created_at DATETIME DEFAULT CURRENT_TIMESTAMP, -- Waktu dibuat

    FOREIGN KEY (role_id) REFERENCES Roles(role_id) -- Relasi ke Roles
);

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-- BAGIAN 4: MEMBUAT TABEL SUPPLIERS

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CREATE TABLE Suppliers (
    supplier_id INT PRIMARY KEY AUTO_INCREMENT, -- ID supplier
    supplier_code VARCHAR(20) UNIQUE NOT NULL, -- Kode supplier
    name VARCHAR(100) NOT NULL, -- Nama supplier
    contact_person VARCHAR(100), -- Kontak
    phone VARCHAR(20), -- Telepon
    email VARCHAR(100), -- Email
    address TEXT, -- Alamat
    is_active BOOLEAN DEFAULT 1, -- Status aktif
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP, -- Dibuat
    updated_at DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP -- Update otomatis
);
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-- BAGIAN 5: MEMBUAT TABEL PRODUCTS

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CREATE TABLE Products ( -- Data produk
    product_id INT PRIMARY KEY AUTO_INCREMENT, -- ID produk
    barcode VARCHAR(50) UNIQUE, -- Barcode
    name VARCHAR(100) NOT NULL, -- Nama produk
```

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price DECIMAL(15,2) NOT NULL, -- Harga
stock INT NOT NULL DEFAULT 0, -- Stok
min_stock INT DEFAULT 5, -- Minimum stok
supplier_id INT, -- Relasi supplier
created_by INT, -- Dibuat oleh user
created_at DATETIME DEFAULT CURRENT_TIMESTAMP, -- Waktu buat
updated_at DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP, -- Update
FOREIGN KEY (supplier_id) REFERENCES Suppliers(supplier_id), -- Relasi supplier
FOREIGN KEY (created_by) REFERENCES Users(user_id) -- Relasi user
);

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-- BAGIAN 6: MEMBUAT TABEL ORDERS

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CREATE TABLE Orders ( -- Data transaksi
    order_id INT PRIMARY KEY AUTO_INCREMENT, -- ID order
    invoice_no VARCHAR(20) UNIQUE NOT NULL, -- Nomor invoice
    user_id INT NOT NULL, -- Kasir
    customer_name VARCHAR(100) DEFAULT 'Umum', -- Nama customer
    subtotal DECIMAL(15,2) NOT NULL, -- Subtotal
    discount DECIMAL(15,2) DEFAULT 0, -- Diskon
    tax DECIMAL(15,2) DEFAULT 0, -- Pajak
    total DECIMAL(15,2) NOT NULL, -- Total
    payment DECIMAL(15,2) NOT NULL, -- Bayar
    change_due DECIMAL(15,2) NOT NULL, -- Kembalian
    status ENUM('completed', 'cancelled', 'refund') DEFAULT 'completed', -- Status

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        created_at DATETIME DEFAULT CURRENT_TIMESTAMP, -- Waktu transaksi
        FOREIGN KEY (user_id) REFERENCES Users(user_id) -- Relasi user
    );

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-- BAGIAN 7: MEMBUAT TABEL ORDER_ITEMS

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CREATE TABLE Order_Items ( -- Detail transaksi
    item_id INT PRIMARY KEY AUTO_INCREMENT, -- ID item
    order_id INT NOT NULL, -- ID order
    product_id INT NOT NULL, -- ID produk
    qty INT NOT NULL, -- Jumlah
    price DECIMAL(15,2) NOT NULL, -- Harga
    total DECIMAL(15,2) NOT NULL, -- Total harga
    FOREIGN KEY (order_id) REFERENCES Orders(order_id) ON DELETE CASCADE, --
    Hapus otomatis
    FOREIGN KEY (product_id) REFERENCES Products(product_id) -- Relasi produk
);

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-- BAGIAN 8: MEMBUAT TABEL ACTIVITY_LOGS

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CREATE TABLE Activity_Logs ( -- Log aktivitas user
    log_id INT PRIMARY KEY AUTO_INCREMENT, -- ID log

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user_id INT, -- User
action VARCHAR(50) NOT NULL, -- Aksi
description TEXT, -- Deskripsi
ip_address VARCHAR(45), -- IP
created_at DATETIME DEFAULT CURRENT_TIMESTAMP, -- Waktu
FOREIGN KEY (user_id) REFERENCES Users(user_id) -- Relasi user
);
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-- BAGIAN 9: MEMBUAT TRIGGER

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DELIMITER //
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CREATE TRIGGER trg_reduce_stock -- Trigger kurangi stok saat transaksi
AFTER INSERT ON Order_Items
FOR EACH ROW
BEGIN
    UPDATE Products SET stock = stock - NEW.qty
    WHERE product_id = NEW.product_id;
END//
```

```
CREATE TRIGGER trg_restore_stock -- Trigger kembalikan stok saat cancel
AFTER UPDATE ON Orders
FOR EACH ROW
BEGIN
    IF NEW.status = 'cancelled' AND OLD.status = 'completed' THEN
        UPDATE Products p
```

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        JOIN Order_Items oi ON p.product_id = oi.product_id

        SET p.stock = p.stock + oi.qty

        WHERE oi.order_id = NEW.order_id;

    END IF;

END//

DELIMITER ;

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-- BAGIAN 10: MEMBUAT INDEX

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CREATE INDEX idx_orders_user_id ON Orders(user_id); -- Index pencarian user
CREATE INDEX idx_orders_created_at ON Orders(created_at); -- Index tanggal
CREATE INDEX idx_products_barcode ON Products(barcode); -- Index barcode
CREATE INDEX idx_products_supplier ON Products(supplier_id); -- Index supplier
CREATE INDEX idx_order_items_order ON Order_Items(order_id); -- Index order
CREATE INDEX idx_order_items_product ON Order_Items(product_id); -- Index produk

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-- BAGIAN 11: INSERT DATA ROLES

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INSERT INTO Roles (role_id, role_name, level) VALUES -- Memasukkan data role
(1, 'admin', 5),      -- Admin: akses penuh ke sistem
(2, 'manager', 4),    -- Manager: bisa laporan dan manajemen user

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(3, 'senior_cashier', 3), -- Kasir senior: bisa transaksi dan laporan terbatas
(4, 'junior_cashier', 2), -- Kasir junior: hanya transaksi
(5, 'owner', 5);      -- Owner: akses penuh seperti admin
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-- BAGIAN 12: INSERT DATA USERS

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INSERT INTO Users (user_id, username, password, full_name, role_id, is_active) VALUES
-- Data user
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(1, 'admin', 'admin123', 'Budi Santoso', 1, 1), -- User admin
(2, 'kasir_senior', 'senior123', 'Siti Aisyah', 3, 1), -- Kasir senior
(3, 'kasir_junior', 'junior123', 'Andi Wijaya', 4, 1), -- Kasir junior
(4, 'manager', 'manager123', 'Dian Permata', 2, 1), -- Manager
(5, 'owner', 'owner123', 'Hendra Gunawan', 5, 1); -- Owner
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-- BAGIAN 13: INSERT DATA SUPPLIERS

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INSERT INTO Suppliers (supplier_code, name, contact_person, phone, email, address)
VALUES -- Data supplier
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('SP001', 'CV. Sumber Makmur', 'Budi Santoso', '081234567890',
'sumbermakmur@email.com', 'Jakarta'), -- Supplier 1
('SP002', 'PT. Indofood Sukses', 'Ani Wijaya', '081298765432', 'indofood@email.com',
'Surabaya'), -- Supplier 2
```


('SP003', 'Toko Aqua Sentosa', 'Cahyo P', '085678901234', 'aqua@email.com', 'Bandung'), -- Supplier 3

('SP004', 'Distro Roti Sari Roti', 'Dewi Lestari', '087812345678', 'sariroti@email.com', 'Medan'); -- Supplier 4

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-- BAGIAN 14: INSERT DATA PRODUCTS

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INSERT INTO Products (product_id, barcode, name, price, stock, min_stock, supplier_id, created_by) VALUES -- Data produk

(1, '8991001100010', 'Indomie Goreng', 3500, 150, 10, 2, 1), -- Produk mie

(2, '8991001100027', 'Indomie Kuah', 3500, 120, 10, 2, 1),

(3, '8998866201234', 'Aqua 600ml', 4000, 200, 20, 3, 1), -- Minuman

(4, '8998866201241', 'Teh Botol Sosro', 5000, 80, 5, 3, 1),

(5, '8999999123456', 'Pocky Stik', 8000, 45, 5, NULL, 1), -- Tanpa supplier

(6, '8991002100033', 'Mie Sedap Goreng', 3200, 100, 10, 2, 1),

(7, '8991002100040', 'Mie Sedap Kuah', 3200, 90, 10, 2, 1),

(8, '8998866201258', 'Fruit Tea', 4500, 60, 5, 3, 1),

(9, '8999999888888', 'Roti Tawar', 12000, 30, 3, 4, 1),

(10, '8991005000010', 'Gula Pasir 1kg', 15000, 50, 5, 1, 1),

(11, '8991006000020', 'Minyak Goreng 1L', 18000, 40, 5, 1, 1),

(12, '8991007000030', 'Telur Ayam 1kg', 28000, 25, 3, 1, 1);

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-- BAGIAN 15: INSERT DATA ORDERS

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INSERT INTO Orders (order_id, invoice_no, user_id, customer_name, subtotal, discount, tax, total, payment, change_due, status, created_at) VALUES -- Data transaksi

(1, 'INV-001', 2, 'Budi', 15000, 0, 0, 15000, 20000, 5000, 'completed', '2026-04-26 10:00:00'),
-- Transaksi 1

(2, 'INV-002', 3, 'Ani', 19400, 1000, 0, 18400, 20000, 1600, 'completed', '2026-04-26 11:00:00'),

(3, 'INV-003', 2, 'Caca', 33000, 2000, 0, 31000, 50000, 19000, 'completed', '2026-04-27 09:00:00'),

(4, 'INV-004', 3, 'Umum', 8500, 0, 0, 8500, 10000, 1500, 'completed', '2026-04-27 10:00:00'),

(5, 'INV-005', 2, 'Rina', 27000, 0, 0, 27000, 50000, 23000, 'completed', '2026-04-27 13:00:00'),

(6, 'INV-006', 3, 'Joko', 18000, 0, 0, 18000, 20000, 2000, 'completed', '2026-04-27 14:00:00'),

(7, 'INV-007', 2, 'Dewi', 61000, 2000, 0, 59000, 100000, 41000, 'completed', '2026-04-27 16:00:00'),

(8, 'INV-008', 3, 'Tono', 17600, 1000, 0, 16600, 20000, 3400, 'completed', '2026-04-28 08:00:00'),

(9, 'INV-009', 2, 'Sari', 35000, 2000, 0, 33000, 50000, 17000, 'completed', '2026-04-28 10:00:00');

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-- BAGIAN 16: INSERT DATA ORDER_ITEMS

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INSERT INTO Order_Items (item_id, order_id, product_id, qty, price, total) VALUES --
Detail barang tiap transaksi

(1, 1, 1, 2, 3500, 7000), -- Item transaksi 1

(2, 1, 3, 2, 4000, 8000),

(3, 2, 2, 1, 3500, 3500),

(4, 2, 4, 2, 5000, 10000),

(5, 2, 6, 2, 3200, 6400),

(6, 3, 1, 5, 3500, 17500),

(7, 3, 10, 1, 15000, 15000),

(8, 4, 3, 1, 4000, 4000),

(9, 4, 8, 1, 4500, 4500),

(10, 5, 1, 2, 3500, 7000),

(11, 5, 3, 2, 4000, 8000),

(12, 5, 9, 1, 12000, 12000),

(13, 6, 2, 1, 3500, 3500),

(14, 6, 4, 2, 5000, 10000),

(15, 6, 8, 1, 4500, 4500),

(16, 7, 10, 1, 15000, 15000),

(17, 7, 11, 1, 18000, 18000),

(18, 7, 12, 1, 28000, 28000),

(19, 8, 6, 2, 3200, 6400),

(20, 8, 7, 1, 3200, 3200),

(21, 8, 5, 1, 8000, 8000),

(22, 9, 1, 3, 3500, 10500),

(23, 9, 3, 2, 4000, 8000),

(24, 9, 9, 1, 12000, 12000),

(25, 9, 8, 1, 4500, 4500);

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-- BAGIAN 17: INSERT DATA ACTIVITY_LOGS

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INSERT INTO Activity_Logs (user_id, action, description, ip_address) VALUES -- Log aktivitas

(1, 'login', 'Admin login', '192.168.1.10'),

(2, 'add_transaction', 'Transaksi INV-001', '192.168.1.12'),

(3, 'add_transaction', 'Transaksi INV-002', '192.168.1.14');

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-- BAGIAN 18: CRUD (CREATE, READ, UPDATE, DELETE)

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INSERT INTO Suppliers (supplier_code, name, contact_person, phone, email, address)

VALUES ('SP005', 'UD. Sumber Rezeki', 'Wahyu Putra', '081377788899', 'sumberrezeki@email.com', 'Semarang');

-- CREATE: menambahkan data supplier baru ke tabel Suppliers

INSERT INTO Products (barcode, name, price, stock, min_stock, supplier_id, created_by)

VALUES ('8999999999999', 'Cokelat Batang', 12000, 50, 5, 5, 1);

-- CREATE: menambahkan produk baru beserta harga dan stok

INSERT INTO Users (username, password, full_name, role_id, is_active)

VALUES ('kasir_baru', 'baru123', 'Rina Febrianti', 4, 1);

-- CREATE: menambahkan user baru (kasir)

SELECT product_id, name, stock, min_stock, price

FROM Products

WHERE stock <= min_stock; -- Menampilkan stok menipis

SELECT invoice_no, customer_name, total, created_at

FROM Orders

WHERE DATE(created_at) = '2026-04-27'; -- Transaksi tanggal tertentu

UPDATE Products SET price = 16000 WHERE product_id = 10; -- Update harga

UPDATE Products SET stock = stock + 10 WHERE product_id = 5; -- Tambah stok

UPDATE Users SET last_login = NOW() WHERE user_id = 1; -- Update login

START TRANSACTION; -- Mulai transaksi

INSERT INTO Suppliers (supplier_code, name, phone)

VALUES ('SP999', 'Toko Dummy', '08123456789');

-- CREATE sementara (data percobaan)

DELETE FROM Suppliers WHERE supplier_id = @test_id; -- Hapus

ROLLBACK; -- Batalkan

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-- BAGIAN 19: JOIN

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SELECT

o.invoice_no,

-- nomor invoice

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    u.full_name AS cashier,          -- nama kasir dari tabel Users
    o.customer_name,                 -- nama pelanggan
    o.total,                         -- total pembayaran
    o.created_at                     -- waktu transaksi
FROM Orders o                      -- tabel Orders diberi alias "o"
INNER JOIN Users u ON o.user_id = u.user_id;
-- menggabungkan Orders dan Users berdasarkan user_id

SELECT
    p.product_id,                   -- id produk
    p.name AS product_name,         -- nama produk
    p.price,                        -- harga produk
    s.name AS supplier_name,        -- nama supplier
    CASE
        WHEN s.supplier_id IS NULL THEN 'Tidak ada supplier' -- jika tidak ada supplier
        ELSE 'Ada supplier'         -- jika ada supplier
    END AS status
FROM Products p
LEFT JOIN Suppliers s ON p.supplier_id = s.supplier_id;
-- LEFT JOIN: semua produk tampil walaupun tidak punya supplier

```

```

SELECT
    s.supplier_code,
    s.name AS supplier_name,
    COUNT(p.product_id) AS total_product, -- jumlah produk per supplier
    GROUP_CONCAT(p.name SEPARATOR ', ') AS list_products -- daftar produk digabung
    jadi l
FROM Products p
RIGHT JOIN Suppliers s ON p.supplier_id = s.supplier_id
GROUP BY s.supplier_id, s.supplier_code, s.name;
-- RIGHT JOIN: semua supplier tetap tampil walau tidak punya produk

```

```

SELECT
    o.invoice_no,
    u.full_name AS cashier,
    o.customer_name,
    p.name AS product_name,
    oi.qty,
    oi.price,
    s.name AS supplier_name
FROM Order_Items oi
JOIN Orders o ON oi.order_id = o.order_id
JOIN Users u ON o.user_id = u.user_id
JOIN Products p ON oi.product_id = p.product_id
LEFT JOIN Suppliers s ON p.supplier_id = s.supplier_id
LIMIT 10;
-- join banyak tabel untuk detail transaksi

```

```

SELECT
    u.user_id,
    u.full_name AS cashier,
    COUNT(o.order_id) AS total_transaksi, -- jumlah transaksi
    SUM(o.total) AS total_omzet,          -- total pendapatan
    ROUND(AVG(o.total), 2) AS rata_rata_transaksi, -- rata-rata
    MAX(o.total) AS transaksi_terbesar,    -- transaksi terbesar
    MIN(o.total) AS transaksi_terkecil     -- transaksi terkecil
FROM Orders o
JOIN Users u ON o.user_id = u.user_id
GROUP BY u.user_id, u.full_name
ORDER BY total_omzet DESC;
-- laporan performa kasir

```

```

SELECT
    p.product_id,
    p.name AS product_name,
    SUM(oi.qty) AS total_terjual,      -- total barang terjual
    SUM(oi.total) AS total_penjualan,  -- total uang penjualan
    COUNT(DISTINCT oi.order_id) AS jumlah_transaksi
FROM Order_Items oi
JOIN Products p ON oi.product_id = p.product_id
GROUP BY p.product_id, p.name
ORDER BY total_terjual DESC
LIMIT 5;

```

-- top 5 produk terlaris

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-- BAGIAN 20: SUBQUERY

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SELECT name, price
FROM Products
WHERE price > (SELECT AVG(price) FROM Products);
-- ambil produk dengan harga di atas rata-rata

```

```

SELECT
    p.name,
    p.price,
    (SELECT SUM(oi.qty) FROM Order_Items oi WHERE oi.product_id = p.product_id) AS
total_terjual

```


FROM Products p

LIMIT 5;

-- menghitung total terjual per produk dengan subquery

SELECT name, price, stock

FROM Products p

WHERE EXISTS (SELECT 1 FROM Order_Items oi WHERE oi.product_id = p.product_id);

-- ambil produk yang pernah terjual (ada di Order_Items)

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-- BAGIAN 21: QUERY DENGAN CASE

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SELECT

name, -- nama produk

stock, -- jumlah stok

CASE

WHEN stock <= min_stock THEN 'Stok Menipis!'

WHEN stock <= min_stock * 2 THEN 'Stok Sedang'

ELSE 'Stok Aman'

END AS status_stok, -- kategori stok

CASE

WHEN price < 5000 THEN 'Murah'

WHEN price BETWEEN 5000 AND 15000 THEN 'Sedang'

ELSE 'Mahal'

END AS kategori_harga -- kategori harga

FROM Products;

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-- BAGIAN 22: UNION

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```

SELECT name AS nama, 'Supplier' AS tipe FROM Suppliers

UNION

SELECT full_name AS nama, 'User' AS tipe FROM Users

LIMIT 10;

-- menggabungkan data supplier dan user dalam 1 hasil

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-- BAGIAN 23: LAPORAN AKHIR

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SELECT '===== ' AS 'HASIL DATABASE';

-- tampilan header

SELECT 'Jumlah Transaksi:' AS 'INFORMASI', COUNT(*) AS 'TOTAL' FROM Orders;

-- total transaksi

SELECT 'Jumlah Produk:' AS 'INFORMASI', COUNT(*) AS 'TOTAL' FROM Products;

-- total produk

SELECT 'Jumlah User:' AS 'INFORMASI', COUNT(*) AS 'TOTAL' FROM Users;

-- total user

```
SELECT 'Jumlah Supplier:' AS 'INFORMASI', COUNT(*) AS 'TOTAL' FROM Suppliers;  
-- total supplier
```

```
SELECT '===== ' AS ";
```

```
SELECT '5 Transaksi Terbaru:' AS ";
```

```
SELECT order_id, invoice_no, customer_name, total, created_at  
FROM Orders  
ORDER BY order_id DESC  
LIMIT 5;  
-- menampilkan 5 transaksi terbaru
```

```
SELECT '===== ' AS ";
```

```
SELECT 'Total Pendapatan:' AS ", SUM(total) AS 'Rp' FROM Orders;  
-- total seluruh pendapatan
```

```
SELECT '===== ' AS ";
```

```
SELECT 'DATABASE BERHASIL' AS 'STATUS';  
-- status akhir
```

```
SELECT '===== ' AS ";
```